Major Features and Complications of Kawasaki Syndrome

Clinical Features of Acute Kawasaki Syndrome

- **Fever**
- **Acute multisystem vasculitis**
- **Polymorphous exanthem**
- **Bilateral bulbar nonexudative conjunctival injection**
- **Inflammation of the oropharynx**
- **Erythema of the palms or soles, edema of the hands or feet, or periungual desquamation**
- **Acute non-suppurative cervical lymphadenopathy** (≥1.5 cm in diameter)

Complications of Kawasaki Syndrome

- **Coronary artery abnormalities**, including aneurysms
- **Myocarditis**
- **Arthralgia and arthritis**
- **Urethritis**
- **Aseptic meningitis**

Prompt treatment with **intravenous immunoglobulin (IVIG)** and **aspirin (ASA)** significantly reduces the risk of coronary artery abnormalities.

Epidemiology

Nationwide epidemics with wave-like spread occurred in Japan in **1979, 1982**, and **late 1985 to early 1986**. Since the mid-1980s, the incidence of Kawasaki syndrome (KS) in Japan has steadily increased, with the **1998 incidence more than 1.5 times higher than in 1987**. The endemic annual incidence is approximately **111 per 100,000 children under 5 years of age**.

Both endemic and epidemic cases in Japan occur most frequently in **late winter and spring**, with a smaller peak in **late fall**. KS is increasingly recognized in other Asian countries. In **Taiwan (1966–2002)**, the annual incidence was **146 per 100,000**, with a peak during **summer months**.

In the **United States and Europe**, KS occurs **sporadically** or as **community-wide outbreaks**. The highest U.S. incidence is reported in **Hawaii** (hospitalization rate: **47.7 per 100,000 children <5 years**). In the continental U.S., incidence ranges from **9 to 19 per 100,000**.

- **Peak age**: 18 to 24 months
- **>80% of cases** occur in children <**5 years**, with most under **2 years**
- **Infants <12 months**, especially those <**6 months**, are at higher risk for coronary artery abnormalities
- **Prevalence by race**: Highest in **Asians**, intermediate in **Blacks**, lowest in **Whites**
- **Clinical presentation** is similar across all racial groups

Although **antecedent respiratory illness** and **exposure to rug cleaning** have been associated with KS, neither has been consistently confirmed as a risk factor.

Etiology and Pathogenesis

Substantial evidence supports **immune activation** as a central mechanism in Kawasaki syndrome:

1. **Activated immune cells**: The acute phase features increased numbers of **activated helper T cells**, **monocyte/macrophages**, and **polyclonal B-cell activation**.

2. **Proinflammatory cytokines**: Elevated serum levels of **IL-1**, **TNF-α**, **IL-6**, **adrenomedullin**, and **vascular endothelial growth factor** correlate with the development of coronary artery abnormalities.
3. **Histopathology**: Vascular lesions in medium-sized arteries suggest **immune-mediated injury**.
4. **Autoantibodies**: Circulating antibodies cytotoxic to **cardiac myosin** and **IL-1/TNF-α/IFN-γ-stimulated endothelial cells** are present.
5. **Response to therapy**: **IVIG plus aspirin** reduces immune activation; **aspirin alone** does not.

Despite these findings, the **exact etiology remains unknown**. KS is hypothesized to result from an **abnormal immune response to an infectious trigger**, supported by:

- Acute, self-limited course
- Seasonal and geographic clustering
- Predilection for young children

Extensive investigations (serologic, culture, molecular) have **failed to identify a consistent causative agent**. At one time, a **novel human coronavirus** was suspected, but this has not been confirmed.